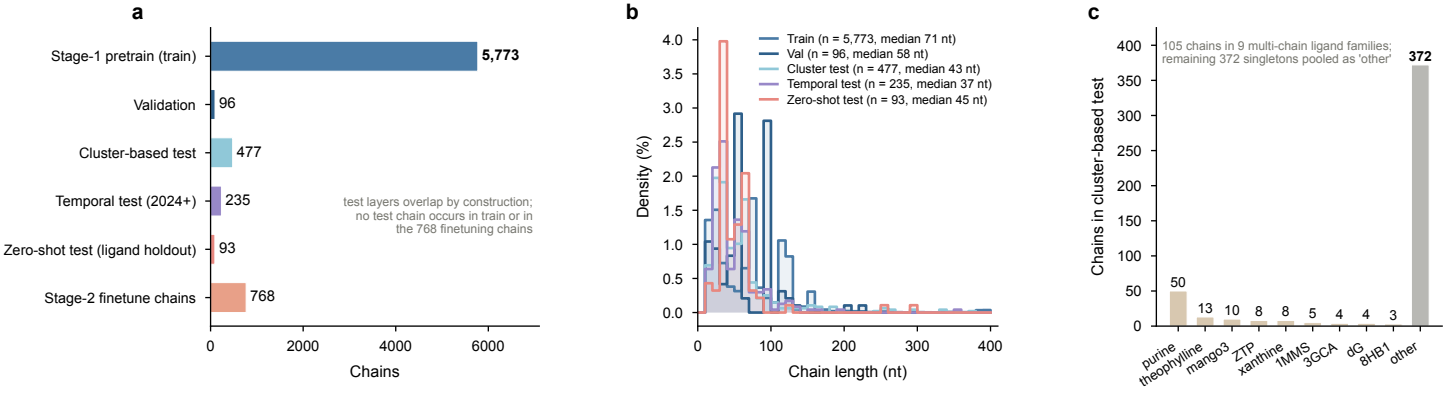
a
Stage-1 pretrain (train)
5,773
Validation
96
Cluster-based test
477
Temporal test (2024+)
235
Zero-shot test (ligand holdout)
93
Stage-2 finetune chains
768
test layers overlap by construction;
no test chain occurs in train or in
the 768 finetuning chains
0
2000
4000
6000
Chains
b
Train (n = 5,773, median 71 nt)
Val (n = 96, median 58 nt)
Cluster test (n = 477, median 43 nt)
Temporal test (n = 235, median 37 nt)
Zero-shot test (n = 93, median 45 nt)
Density (%)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
100
200
300
400
Chain length (nt)
c
105 chains in 9 multi-chain ligand families;
remaining 372 singletons pooled as 'other'
Chains in cluster-based test
0
50
100
150
200
250
300
350
400
50
13
10
8
8
5
4
4
3
372
purine
theophylline
mango3
ZTP
xanthine
1MMS
3GCA
dG
8HB1
other